



Wangdi

Senior Forestry Officer · Biodiversity, Climate & Forest Governance

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Department of Forests & Park Services, Bhutan

2026 CLP Future Conservationist Team Award

Meritorious Promotion, RCSC

Professional Profile

Public-sector forestry, biodiversity, and environmental management professional with five years of progressive experience in Bhutan's forest governance system. Currently serving as Senior Forestry Officer in the NWFP Section, Forest Resources Planning & Management Division, following a merit-based promotion within the first four years of service. Expertise spans biodiversity conservation, climate-responsive forest management, NWFP governance, GIS and remote sensing, ecological research, and policy-linked environmental planning. Team Leader and recipient of the 2026 Conservation Leadership Programme (CLP) Future Conservationist Team Award for *Connected Skies: Bhutan's First GPS-AI Study Empowering Communities to Protect Great Hornbills*. Also appointed as an Adjunct Lecturer at the College of Natural Resources, Royal University of Bhutan, to deliver Forest Mensuration (MEN201).

Career Highlights

MERIT PROMOTION

Promoted to Senior Forestry Officer effective 1 January 2026.

2026 CLP AWARD

Team Leader and recipient of the CLP Future Conservationist Team Award for Connected Skies.

TECHNICAL PLANNING

Led climate vulnerability, human-elephant coexistence, corridor, divisional management, and biodiversity planning outputs.

LANDSCAPE CONSERVATION

IKI Living Landscape focal officer and contributor to Bhutan for Life implementation in Sarpang.

PROGRAMME LEADERSHIP

Managed conservation programmes and annual budgets exceeding Nu. 35 million.

UNIVERSITY TEACHING

Appointed Adjunct Lecturer at CNR, RUB, for Forest Mensuration (MEN201), 75 contact hours.

BIODIVERSITY MONITORING

Established camera-trap monitoring across RMNP, PWS, and Biological Corridor 03, documenting 100+ species.

SPATIAL SCIENCE

Applied GIS, remote sensing, and species distribution modelling to conservation planning and decision-making.

Professional Experience

Senior Forestry Officer

Mar 2026 – Present

NWFP Section, Forest Resources Planning & Management Division · Department of Forests & Park Services

- Administer national-level non-wood forest product (NWFP) allocation processes, utilization permits, quotas, and royalty-related documentation to support equitable resource distribution, sustainable utilization, and regulatory compliance.
- Monitor harvesting and utilization trends and contribute to NWFP management plans, sustainable harvesting guidelines, and national forestry policy and planning processes.
- Support resource assessments and inventories of commercially and ecologically important NWFP species and provide technical guidance on species identification, data collection, sustainable harvesting, and compliance monitoring.
- Promote community-based NWFP management by supporting local groups in sustainable resource use, participatory management, institutional strengthening, and linking forest conservation with rural livelihoods and income-generation opportunities.
- Contribute to the development of a forest-based bioeconomy by supporting sustainable commercialization, value-chain development, value addition, market linkages, and responsible private-sector engagement for NWFPs.
- Support initiatives that connect biodiversity conservation, sustainable forest management, rural enterprise development, and livelihood diversification, particularly for forest-dependent communities.
- Prepare technical inputs for forestry planning documents, annual reports, management guidelines, policy processes, and cross-agency coordination.
- Liaise with divisional forest offices, community groups, government agencies, research institutions, and private-sector stakeholders on NWFP implementation, reporting, sustainable resource management, market development, and field-level issues.

Forestry / Biodiversity Officer — Head, Nature Conservation Section

Divisional Forest Office, Sarpang · Department of Forests & Park Services

Jan 2022 – Feb 2026

- Meritorious promotion by the Royal Civil Service Commission effective 1 January 2026.
- Served as focal officer for the IKI Living Landscape Project and supported Bhutan for Life conservation interventions, including biodiversity monitoring and field coordination.
- Managed annual conservation budgets exceeding Nu. 35 million.
- Established and maintained a camera-trap monitoring network across Royal Manas National Park, Phibsoo Wildlife Sanctuary, and Biological Corridor 03, documenting 100+ species.
- Conducted biodiversity assessments across multiple forest ecosystems and elevational gradients.
- Led preparation of a Conflict-to-Coexistence Strategy for human-elephant conflict mitigation and applied GIS, remote sensing, and species distribution modelling for corridor and habitat planning.
- Developed the Conservation Management Plan for Biological Corridor 03 and prepared the Divisional Forest Office Management Plan for Sarpang Forest Division.
- Led the Climate Change Vulnerability and Capacity Assessment for Sarpang Forest Division.
- Developed the Environmental and Social Safeguards Management Plan for divisional development activities and authored the Biodiversity Status Report for Sarpang Forest Division.

Academic & Teaching Experience

Adjunct Lecturer — Forest Mensuration (MEN201)

Aug 2026 – Dec 2026

College of Natural Resources, Royal University of Bhutan · FO Year 2 · 75 contact hours

- Appointed to deliver Forest Mensuration (MEN201) in accordance with the approved curriculum, academic schedule, and University regulations.
- Responsible for lectures and practical instruction, student attendance and assessment records, examinations and assessments, and submission of required academic documentation.
- Teach core forest mensuration concepts and applied measurement methods for undergraduate Forestry students.

Education

Bachelor of Science (Honours) in Forestry — First Class

College of Natural Resources, Royal University of Bhutan · Overall Grade: 79%

Jul 2017 – Jun 2021

Thesis: Analysis of Physical and Chemical Properties of Natural Salt Licks and Determination of Animal Presence.

Methods: Ecological field surveys, wildlife sign assessment, camera-trap monitoring, soil and mineral analysis, biodiversity data collection, statistical analysis, scientific writing, and interpretation of species-environment relationships.

Selected Conservation & Research Project

Connected Skies: Bhutan's First GPS-AI Study: Empowering Communities to Protect Great Hornbills

Team Leader · 2026 Conservation Leadership Programme Future Conservationist Team Award

- Lead a multidisciplinary conservation project combining GPS-GSM tracking, AI-assisted bioacoustic monitoring, field ecology, and community engagement for Great Hornbill conservation in south-central Bhutan.
- Study movement routes, home ranges, seasonal habitat use, nesting and feeding areas, and potential connectivity bottlenecks to inform evidence-based conservation action.
- Coordinate project implementation, field research, stakeholder engagement, and community-based conservation activities.

Selected Publications

- Wangdi, et al. (2021). Analysis of Physical and Chemical Properties of Natural Salt Licks and Determination of Animal Presence. *Journal of Renewable Natural Resources Bhutan*.
- Wangdi, et al. (2021). Diversity of Aquatic Beetles along Altitudinal Gradients and Water Quality Parameters. *Journal of Renewable Natural Resources Bhutan*.
- Orong, K., Dorjee, S., & Wangdi. (2026). Wild Edible and Medicinal Plants: A Case Study of Community Perceptions, Gendered Roles, and Pathways for Sustainable Management in Yagang and Gengu Villages, Darla Gewog, Chhukha. *Bhutan Journal of Natural Resources & Development*.

Manuscripts in Preparation/Under Review

- Separating Fundamental from Realized Habitat: A Transferable Dual-Model Framework for Quantifying Anthropogenic Constraints on Species Distributions in Human-Dominated Landscapes.
- Vegetation Community Composition and Species-Environment Relationships Along an Elevational Gradient in South-Central Bhutan.
- Climate Refugia Reveal a Lowland Protection Gap and Rising Conflict Exposure for Asian Elephants in Bhutan.
- Climate Change Vulnerability and Adaptive Capacity of Rural Communities in South Central Bhutan: An Integrated Socio-Ecological Assessment.
- Tiger Occupancy and Large Ungulate Distribution Along a Human Pressure Gradient in South-Central Bhutan: A Hierarchical Detection-Occupancy Approach.
- Spatiotemporal Risk Partitioning Among Tigers, Prey, and Human Activity in South-Central Bhutan.
- Community Assembly and Environmental Filtering Along a Disturbance-Climate Gradient in a Himalayan Tiger Landscape.

Technical Expertise

GEOSPATIAL & REMOTE SENSING

QGIS

ArcGIS

Google Earth Engine

Python (GeoPandas & ArcPy)

DATA ANALYSIS & MODELLING

R

Python

SPSS

Machine Learning

Multivariate Analysis

ECOLOGICAL & FIELD METHODS

Species Distribution Modelling

Camera-Trap Analysis

Occupancy Modelling

Climate Modelling

Forest Inventory

Acoustic Monitoring

AI Species ID

PROFESSIONAL STRENGTHS

Project Management

Budget Administration

Team Leadership

Stakeholder Engagement

Scientific & Grant Writing

EIA

Conservation Planning

Climate Adaptation

Professional Development & Certifications

Conservation, Biodiversity & Climate Change

Advanced professional training through ICIMOD, UNDP, and IUCN in climate change adaptation, ecosystem restoration, collaborative conservation planning, biodiversity assessment, and IUCN Red List assessment methodologies.

Remote Sensing, GIS & Earth Observation

Professional training and certifications through NASA ARSET, EO College, and ISRO/NRSC covering satellite remote sensing, GIS-based spatial planning, environmental monitoring, forest-fire detection, active-fire monitoring, and burnt-area mapping using satellite datasets.

Data Science, Geospatial Analytics & Modelling

Applied training in R, Python, QGIS, Google Earth Engine, statistical modelling, machine learning, spatial data analysis, and environmental data visualization, with applications to forestry, biodiversity, and natural-resource management.

Ecological & Conservation Analytics

Specialized training in species distribution modelling, camera-trap data analysis, biodiversity monitoring, ecological data management, and spatial conservation planning.

Emerging Technologies & Digital Applications

Training in drone operations for emergency response, full-stack web development, digital data-management systems, Google Workspace administration, and technology-enabled environmental monitoring.

Sustainability, Economics & Management

Professional development in Environmental, Social and Governance (ESG) principles, production-cost estimation, resource planning, and sustainability-oriented decision-making.

AI & Developer Tools

Anthropic Certificates of Completion in Claude 101 (3 May 2026), Claude Code 101 (3 May 2026), and Claude Code in Action (7 May 2026), covering practical applications of generative AI, AI-assisted coding, and developer workflows.

Honours & Awards

- Conservation Leadership Programme (CLP) Future Conservationist Team Award — 2026 · Team Leader, Connected Skies.
- Meritorious Promotion, Royal Civil Service Commission — effective 1 January 2026.
- SAF-Madanjeet Singh Scholarship Recipient — merit-based full scholarship (2017–2021).

Professional Engagement

- Conservation Practitioner, Bhutan for Life Program.
- Royal Civil Service Commission (RCSC) — public service appointment and career progression framework.

References

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